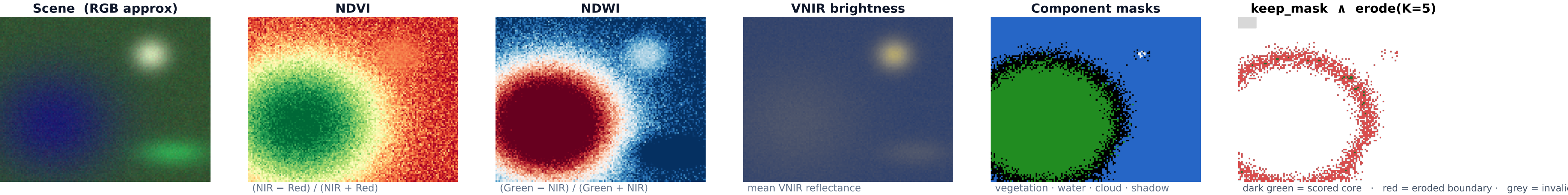


Allotrope · Masking & Ensemble Anomaly Pipeline

Scene segmentation (NDVI / NDWI / brightness) → keep_mask → binary erosion · model fan-out → CDF-normalize → fuse → p99.5 threshold

① Keep-mask construction

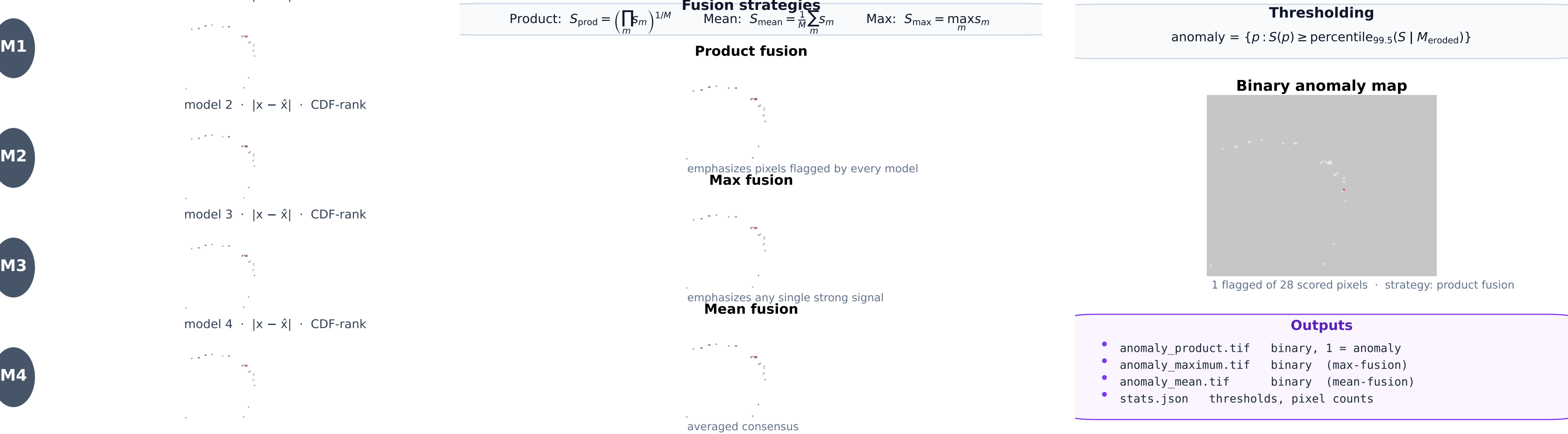
spectral indices → component masks → keep_mask → erosion



eroded keep_mask is passed into every model's score map

② Ensemble anomaly scoring

N reconstruction models in parallel → CDF normalize → fuse via product / max / mean → threshold at p99.5



Fusion strategies

Product: $S_{\text{prod}} = \left(\prod_m s_m\right)^{1/M}$ · Mean: $S_{\text{mean}} = \frac{1}{M} \sum_m s_m$ · Max: $S_{\text{max}} = \max_m s_m$

Product fusion

emphasizes pixels flagged by every model

Max fusion

emphasizes any single strong signal

Mean fusion

averaged consensus

Thresholding

anomaly = $\{p : S(p) \geq \text{percentile}_{99.5}(S \mid M_{\text{eroded}})\}$

Binary anomaly map

1 flagged of 28 scored pixels · strategy: product fusion

Outputs

- anomaly_product.tif binary, 1 = anomaly
- anomaly_maximum.tif binary (max-fusion)
- anomaly_mean.tif binary (mean-fusion)
- stats.json thresholds, pixel counts

each model: forward pass → reconstruction error → mask with eroded keep_mask → CDF-rank to [0, 1]